VMC-3000 Series | MID 144 (VIB) | Bosch CISS Dataset | ISO 10816-3

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1. Introduction

This reference document describes the vibration monitoring system integrated into the VMC-3000 machining centre, the data collected by the Bosch CISS tri-axial accelerometer, the statistical features extracted for condition monitoring, and the ISO 10816-3 severity classification zones used to trigger error codes. This document is the technical basis for the EquipmentIQ predictive maintenance AI system and the RAG-powered diagnostic agent.

2. Vibration Severity — ISO 10816-3 Classification

Machine vibration severity is evaluated per ISO 10816-3 (Mechanical vibration — Evaluation of machine vibration by measurements on non-rotating parts — Part 3: Industrial machines with nominal power above 15 kW and nominal speeds between 120 RPM and 15,000 RPM). The VMC-3000 spindle falls in Group 2 (flexible mounting) of ISO 10816-3.

Zone	RMS Range (mm/s)	Description	Machine Status	Error Codes Triggered
A	0 – 2.3	New machine reference	Optimal — unrestricted long-term operation	VIB-AD-001, VIB-WN-001
B	2.3 – 4.5	Acceptable for long-term	Normal production — monitor trend	VIB-NC-001, VIB-MN-001
B upper	4.5 – 7.1	Elevated but not alarming	Reduce load; inspect tool	VIB-SR-001, VIB-MD-001
C	7.1 – 11.2	Unsatisfactory — short-term only	Plan maintenance within 4 h	VIB-MJ-001, SPN-MJ-002
D	> 11.2	Dangerous — machine damage risk	Emergency stop	VIB-CR-001, SPN-CR-001

Table 2-1: ISO 10816-3 Vibration Severity Zones — VMC-3000

3. Statistical Features Extracted

The EquipmentIQ edge gateway processes raw accelerometer data from the Bosch CISS sensor in real time, extracting eight statistical features per axis per operation cycle. These features form the feature vector used by the predictive maintenance ML model and are stored in the EquipmentIQ time-series database.

Feature	Symbol	Formula	Fault Sensitivity	Threshold Action
RMS	x_rms	`sqrt(mean(x^2))`	Overall energy — rises with any fault	ISO zone classification
Peak	x_peak	`max(|x|)`	Impact events — transient faults	CR threshold if >2x normal
Crest Factor	x_crest	`x_peak / x_rms`	Impulsive faults (early bearing)	Alert if >4.5
Kurtosis	x_kurt	`E[(x-mu)^4] / sigma^4`	Best early-fault indicator	Warning if >5.0, Alarm if >8.0
Mean	x_mean	`mean(x)`	DC offset — static load	Cross-check only
Std Dev	x_std	`std(x)`	Signal spread — general health	Trend monitoring
Min	x_min	`min(x)`	Negative peak value	Absolute limit check
Max	x_max	`max(x)`	Positive peak value	Absolute limit check

Table 3-1: Statistical Features — Per-Axis Extraction (Bosch CISS Data)

4. Fault Category Signatures

The following table describes the characteristic vibration signatures for each fault category observed in the Bosch CNC Machining dataset (3 machines, 15 operations, 2019–2021). Fault categories are used to route queries to the correct diagnostic agent in the EquipmentIQ RAG system.

Fault Category	Dominant Feature	Typical Values (Fault)	Affected Operations	Primary Error Codes
tool_wear	x_rms, spindle load P002	RMS 3–6 mm/s; Kurtosis 3–6	OP01,02,07,08,11	SPN-MJ-003, SPN-SR-001
spindle_bearing_fault	x_crest, x_kurtosis, P004	Crest >5; Kurtosis >8; RMS 4–11 mm/s	OP04,05,10,14	SPN-CR-001, VIB-MJ-001
chatter_vibration	x_rms all axes, P063	RMS 5–9 mm/s; periodic peaks at tooth-pass freq	OP03,06,09	SPN-MJ-004, AXS-SR-001
actuator_fault	P016 following error, servo current	Following error >0.5 mm; current spike	OP00,12,14	AXS-CR-001, TCS-CR-001
process_anomaly	P001 deviation, P083 override	Speed deviation >50 RPM; override ≠100%	Any	CNC-MJ-001, SPN-MD-002

Table 4-1: Fault Category Vibration Signatures — Bosch CNC Dataset

5. Vibration Monitoring Parameters

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P004	4	Spindle Bearing Vibration	mm/s	0 – 4.5	0 – 11.2
P060	60	X-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2
P061	61	Y-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2
P062	62	Z-Axis Vibration RMS	mm/s	0 – 4.5	0 – 11.2
P063	63	Vibration Crest Factor	-	1.0 – 4.5	0 – 12.0
P064	64	Kurtosis Index	-	2.5 – 5.0	0 – 25.0

Table 5-1: VIB Monitored Parameters

6. Vibration Error Codes — Full Reference

Error Code	Severity	Title	Primary Param	Action
VIB-CR-001	CRITICAL	Vibration — ISO Zone D — Machine Stop	—	Immediate machine stop. Do not restart wi
VIB-MJ-001	MAJOR	Vibration — ISO Zone C — Major Alert	—	Stop current operation. Machine must be in
VIB-SR-001	SERIOUS	Vibration — ISO Zone B Upper — Elevated	—	Complete current cycle then stop. Schedul
VIB-MD-001	MODERATE	Crest Factor — Elevated Above Baseline	—	Log fault. Schedule maintenance within 24
VIB-MN-001	MINOR	Kurtosis Index — Rising Trend	—	Log fault. Monitor parameters. Schedule in
VIB-WN-001	WARNING	Vibration — Normal Operating Range	—	Log event. No immediate action required. F
VIB-NC-001	NOTICE	Crest Factor — Within Normal Band	—	Informational. Parameter approaching limit
VIB-AD-001	ADVISORY	Vibration Baseline Update	—	Performance advisory only. No maintenanc

Table 6-1: Vibration Monitoring Error Codes

7. Condition Monitoring — Dataset Reference

The EquipmentIQ condition monitoring system is grounded in real sensor data from the Bosch CNC Machining Dataset (CC-BY-4.0). Dataset characteristics used to train the predictive models and calibrate alarm thresholds:

Parameter	Value
Dataset Source	Bosch Research — CNC_Machining (GitHub)
Citation	Tnani et al. Procedia CIRP 2022, 107, 131-136
License	Creative Commons CC-BY-4.0
Total Files	1,702 HDF5 recordings
Machines	3 (M01, M02, M03)
Operations	15 (OP00–OP14)
Date Range	February 2019 – August 2021
Normal Samples	1,632 (95.9%)
Fault Samples	70 (4.1%)
Fault Categories	5: tool_wear (34), spindle_bearing_fault (20), actuator_fault (8), chatter_vibration (7), process_anomaly
Sampling Rate	2,000 Hz per axis (X, Y, Z)
Sensor	Bosch CISS tri-axial MEMS accelerometer (±8g)
Features Extracted	8 per axis: mean, std, RMS, peak, crest factor, kurtosis, min, max

Table 7-1: Bosch CNC Machining Dataset — Reference Summary